

# Hypo-Stage: Tool-Supported Architectural Hypothesis Engineering in Professional Software Development Teams — A Case Study

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## Master's Thesis Defense

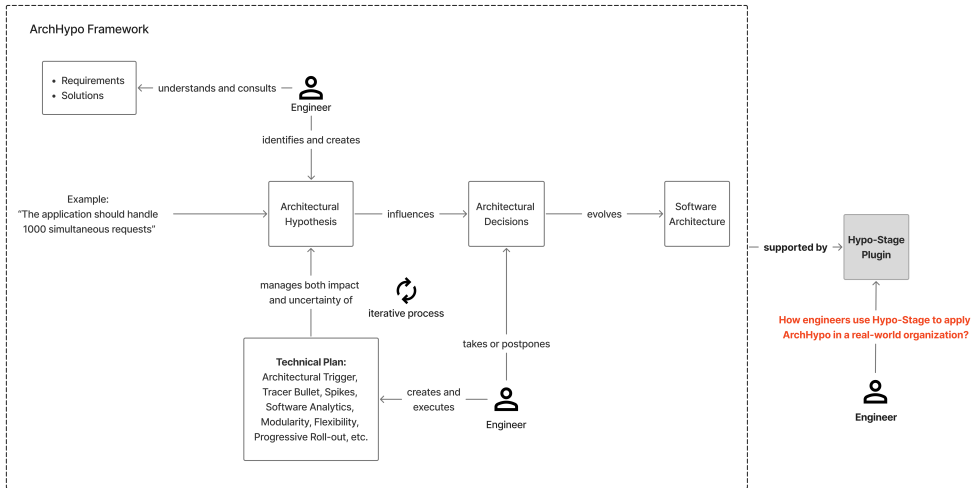
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# Context



# Problem & research questions

## The gap

Architectural uncertainty is unavoidable, yet it stays **implicit** and locked in a few experts

(Silva, Melegati, Silveira, et al., 2025; Souza et al., 2019).

ArchHypo makes it explicit — but teams find it **hard to learn**

(Silva, Melegati, Silveira, et al., 2025; Silva, Melegati, Wang, et al., 2024).

**Hypo-Stage** already provides tool support, yet there is **no real-world usage** to observe its practice.

**RQ1.** How do teams *use* Hypo-Stage and ArchHypo on a scaled product?

**RQ2.** What *uncertainties* and recurring *patterns* do they record?

**RQ3.** What *benefits and challenges* do team members perceive?

**RQ4.** Which *refinements* to the tool emerge, and how do they address the challenges?

# Method & study setup

Single organization · scaled product · Backstage IDP

**Product team**

12 engineers

**Platform team**

4 engineers

**Five 2-week sprints (≈9 weeks)**

Sprint 1

Sprint 2

Sprint 3

Sprint 4

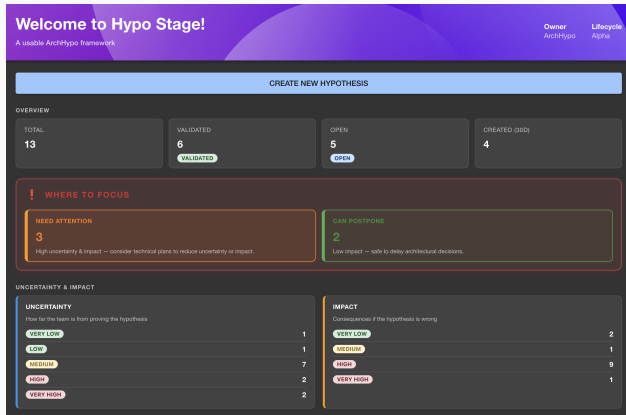
Sprint 5

Participant  
observation

Artifact analysis  
(13 hypotheses)

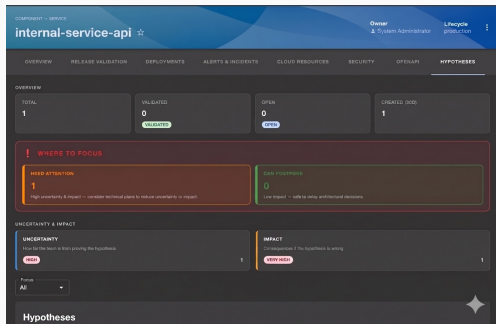
Active-listening  
sessions

Three triangulated data sources



Adoption was **senior-led**  
and **event-anchored**:

- 12 of 13 hypotheses authored by senior/staff engineers
- Registration clustered around protected technical sessions
- Backstage integration cut context-switching cost

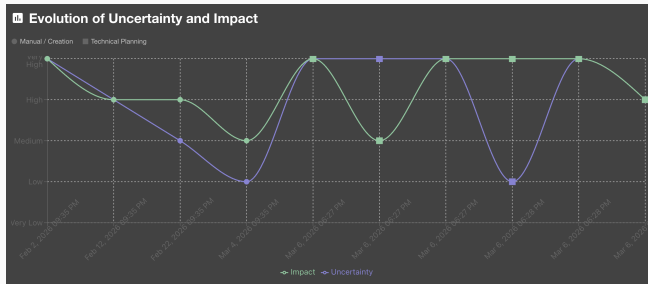


Uncertainty  $\times$  impact, surfaced per service in the developer portal. Dominant quality attributes: **Reliability** and **Auditability**.

## Five emerging hypothesis patterns

- **Hypothesis Cluster** — one platform concern  $\rightarrow$  linked hypotheses
- **Regulatory Trigger** — compliance obligation as tracked uncertainty
- **Context-Agnostic Restructuring** — whether/how to decouple boundaries
- **Load-Validated** — proven under load, implementation still open
- **Shared Resource Governance** — ownership rules for shared infrastructure

First empirically grounded answer to *what shapes* architectural hypotheses take in practice.



Evolution chart: uncertainty & impact tracked across successive assessments.

**+ Create New Hypothesis**

Define your hypothesis and assess its uncertainty and potential impact

Entity References

Please select at least one entity reference. Type to search existing components.

Hypothesis Statement\*

8/500 characters

Source Type\*

**Uncertainty Level**

How far is the team from validating/falsifying this hypothesis?

1 2 3 4 5 Not rated

**Impact Level**

What is the potential impact on the system if the hypothesis proves false?

1 2 3 4 5 Not rated

**Quality Attributes**

Quality Attributes (select one or more)

Please select at least one quality attribute that this hypothesis affects. Click the field to open the list; each option shows a short concept in parentheses.

Inline formulation guidance in the creation form lowers the falsifiability barrier.

**9 refinement categories** — including prioritization, the evolution chart, and form usability — each traced to an observed friction point.

# Implications & lessons learned

## Finding

Even with tool support, hypothesis work stayed **senior-anchored** (12 of 13).

## Implication

A tool removes **friction** but cannot supply the **organizational conditions** for broad participation.

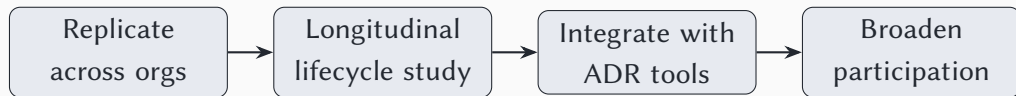
## Lesson

Ship hypothesis support **inside** the developer portal — *extend* the workflow, don't add a new one.

Distributed architectural work needs lightweight structure that preserves team autonomy while protecting shared decision quality.



## Future work & closing



### Takeaway

Tool-mediated ArchHypo succeeds at **integration and traceability**; the remaining challenge is **organizational and pedagogical**.

*Limitations: single organization, 13 hypotheses,  $\approx 9$  weeks, researcher as participant — emerging propositions, not consolidated findings.*